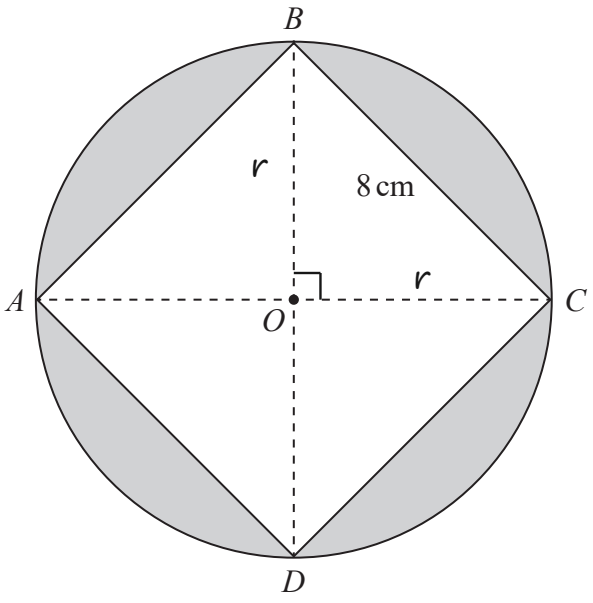
26 The diagram shows a square *ABCD* of side 8 cm inside a circle, centre *O*.
The vertices of the square lie on the circle.



Work out the total area of the four shaded segments.

Give your answer correct to 3 significant figures.

Right triangle BOC: $r^2 + r^2 = 8^2$.

1 mark
Pythagoras

$2r^2 = 64$, so $r^2 = 32$.

1 mark
$r^2=32$

Circle area = $\pi r^2 = 32\pi$ cm^2.

1 mark
Find circle area

Shaded area = circle area – square area.

1 mark
Subtract square64

$32\pi - 8^2 = 36.530964...$ cm^2.

36.5 cm^2 (3 significant figures).

1 mark
36.5cm^2

.....cm^2

(Total for Question 26 is 5 marks)